

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What is the ONE villus-shape difference that separates duodenum, jejunum, and ileum on a slide?
2. Name the 6 cell types found lining the intestinal crypts (Crypts of Lieberkuhn).
3. What do Paneth cell granules release, and where in the crypt are Paneth cells located?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Small intestine — overview

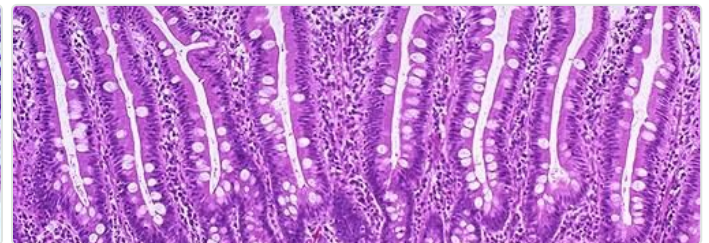
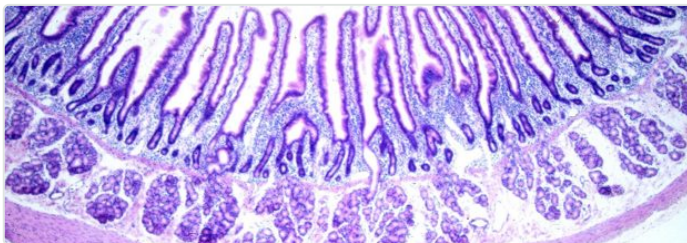
Region	Length
Duodenum	~25 cm — C-shaped, receives bile & pancreatic duct
Jejunum	~2.5 m
Ileum	~3.5 m — ends at ileocecal valve

Total ~5 m. Main functions: **absorption & secretion of digestive enzymes.**

2 Regional identification — the core discriminator

Exam skill: given an unlabelled slide, one villus shape + one extra feature identifies the region.

Feature	Duodenum	Jejunum	Ileum
Villi shape	Leaf-shaped	Long, finger-shaped	Fewer, club-shaped
Goblet cells	Few	Intermediate	Abundant
Plicae circulares	Present	Most numerous / tallest	Present, fewer
Unique landmark	Brunner's glands (submucosa) — alkaline mucus, buffers acidic chyme to optimum pH for pancreatic enzymes	— none distinctive —	Peyer's patches (lamina propria) — lymphoid nodules; overlain by M cells
Serosa/adventitia	Serosa & adventitia	Serosa	Serosa



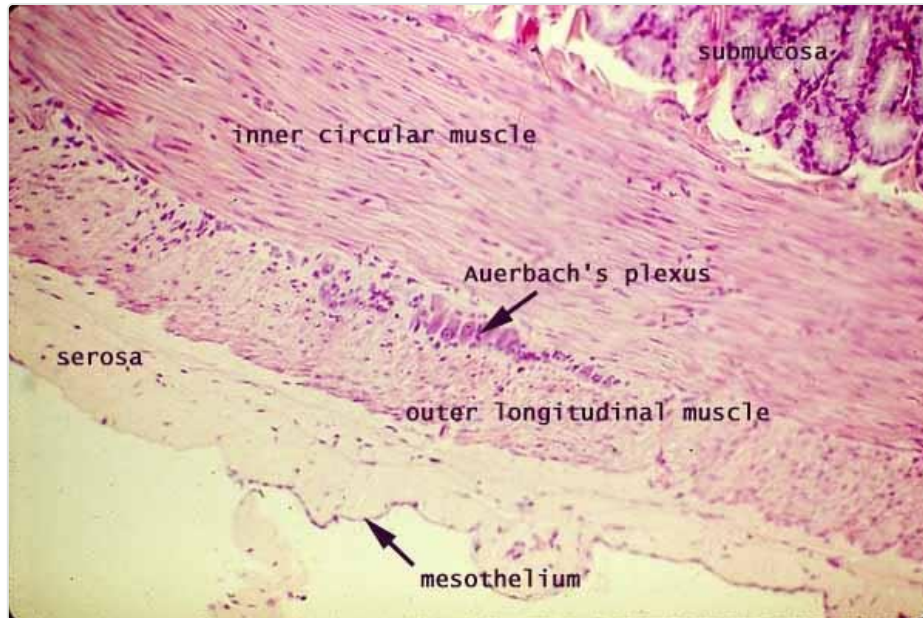
Left: duodenum — leaf-shaped villi. Right: jejunum — long finger-shaped villi.



Ileum — club-shaped villi with a Peyer's patch (lymphoid nodule) bulging into the submucosa.

3 Wall layers — same 4 as the rest of the GI tract

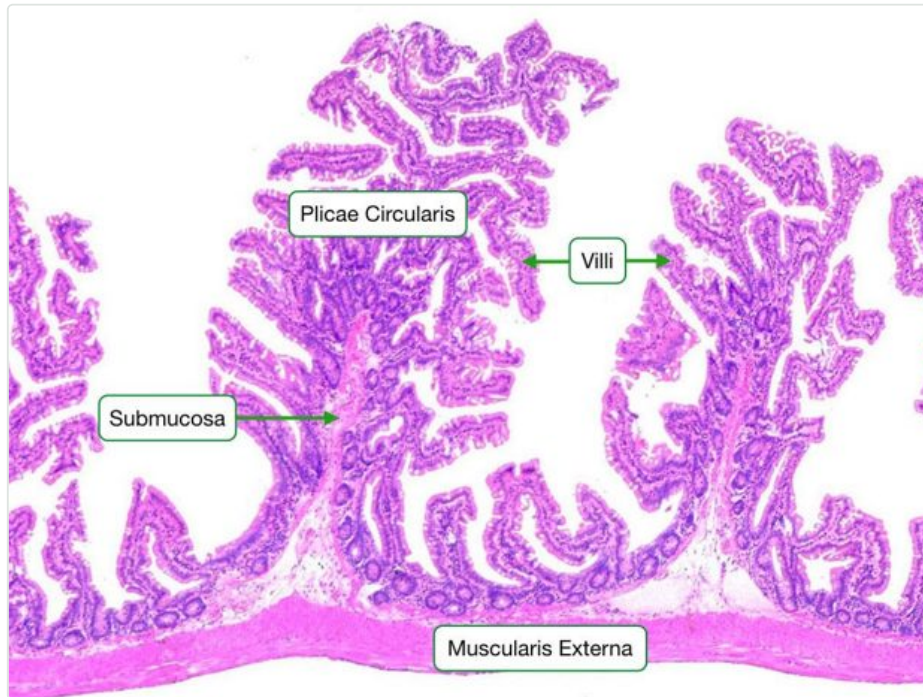
Mucosa	Epithelium + lamina propria (loose c.t. with vessels, nerves, smooth muscle, lymphoid cells, macrophages) + muscularis mucosae
Submucosa	Dense c.t.; contains Brunner's glands only in duodenum
Muscularis externa	Inner circular + outer longitudinal smooth muscle; Auerbach's (myenteric) plexus between the two layers
Serosa / adventitia	Outer covering — mesothelium where intraperitoneal



Auerbach's plexus sits between the inner circular and outer longitudinal muscle layers.

4 How the mucosa maximises surface area

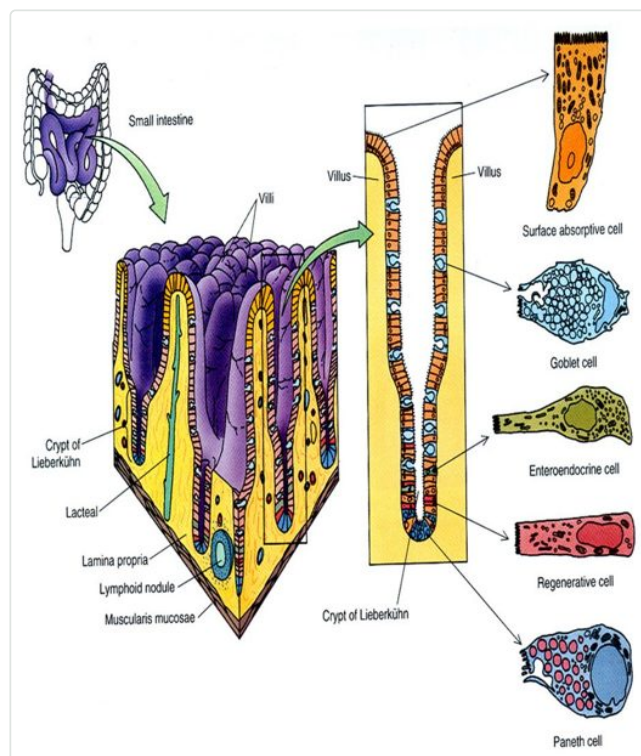
1. Length	~5 m of tract alone increases contact time and area
2. Plicae circulares	Permanent folds of mucosa + submucosa, most prominent in jejunum; slow chyme transit → more time for absorption
3. Villi	Finger-like mucosal projections; core of reticular c.t. with blood capillaries + a central lymphatic (lacteal)
4. Microvilli	Apical cytoplasmic extensions on absorptive cells — the "brush border"



Plicae circulares, villi and muscularis externa in one section.

5 Intestinal glands (Crypts of Lieberkuhn) — cell ID table

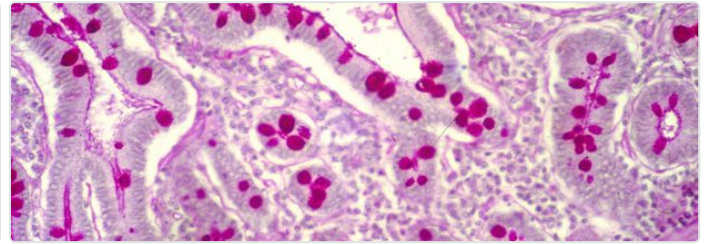
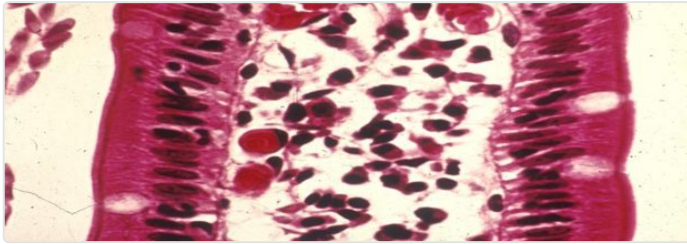
Simple tubular glands extending into the lamina propria below the villus base. Secrete intestinal juice through small openings between villi.



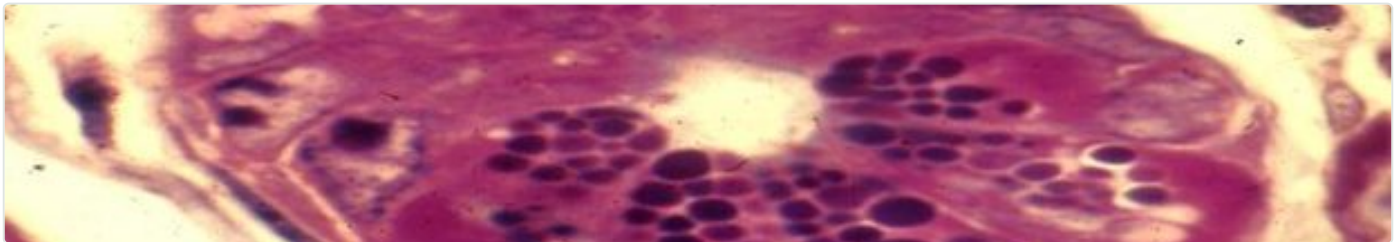
All six crypt/villus cell types in one diagram — use this as the anchor.

Cell type	Location	Key ID feature	Function
Absorptive (columnar) cell	Covers villi & crypts	Tall columnar, oval basal nucleus, brush border of ~3,000 microvilli/cell	Absorption
Goblet cell	Interspersed with absorptive cells; increase toward ileum	Mucin-distended, goblet-shaped	Mucus — protection & lubrication

Paneth cell	Base of crypt	Basophilic cytoplasm, eosinophilic apical granules	Granules release lysozyme, phospholipase A2, defensins — break down microbial membranes; regulate crypt microenvironment
Enteroendocrine cell	Lower third of crypt	Triangular; APUD-type (as in stomach)	Secrete serotonin, somatostatin, CCK, secretin — regulate motility, secretion, appetite
Stem cell	Base of crypt	Undifferentiated, mitotic	Precursor of all other crypt/villus cells; GI epithelium turns over every 2–4 days
M (microfold) cell	Overlies lymphoid follicles (Peyer's patches) — ileum	Discontinuous basement membrane	Endocytoses antigens → transports to underlying macrophages/lymphocytes; protective/immune surveillance



Left: absorptive cell brush border. Right: goblet cells (PAS-type stain).



Paneth cells — basophilic cytoplasm with bright eosinophilic apical granules, at the crypt base.

P Must-remember pearls

- **Duodenum** = leaf-shaped villi + Brunner's glands (submucosa) — buffers acidic chyme.
- **Jejunum** = tallest/most numerous plicae circulares + long finger-shaped villi.
- **Ileum** = club-shaped villi + Peyer's patches (lamina propria) + M cells — the immune surveillance segment.
- **Paneth cells** are the only crypt cell with antimicrobial granules (lysozyme, phospholipase A2, defensins).
- Absorptive cells carry ~3,000 microvilli each — the main site of terminal digestion and absorption.

Q Test yourself

1. A biopsy shows leaf-shaped villi, few goblet cells, and glands in the submucosa secreting alkaline mucus. This segment is the:

- A) Jejunum
B) Ileum
C) Duodenum
D) Cecum

2. Club-shaped villi with abundant goblet cells and a lymphoid nodule bulging from the submucosa localise the segment to the:

- A) Duodenum
B) Jejunum
C) Ileum
D) Colon

3. Which crypt cell releases lysozyme and defensins to help regulate the crypt's microenvironment?

- A) Paneth cell
B) Goblet cell
C) M cell
D) Enteroendocrine cell

4. A specialised epithelial cell overlying a Peyer's patch, with a discontinuous basement membrane, that transports antigens to underlying macrophages is the:

- A) Absorptive cell
B) M cell
C) Paneth cell
D) Stem cell

5. The tallest, most numerous plicae circulares combined with long finger-shaped villi are most characteristic of the:

A) Duodenum
C) Ileum

B) Jejunum
D) Rectum

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. C — Duodenum** — leaf-shaped villi + Brunner's glands are diagnostic.
- 2. C — Ileum** — club-shaped villi + Peyer's patches (lamina propria lymphoid nodules).
- 3. A — Paneth cell** — basophilic cytoplasm, eosinophilic apical granules at crypt base.
- 4. B — M cell** — overlies Peyer's patches; discontinuous basement membrane; antigen transport.
- 5. B — Jejunum** — longest segment with the most prominent plicae circulares.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Small Intestine SGL2, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.